Graph Construction from a Statistical Vocabulary

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Abstract

This extended abstract presents work in preparation that applies our group's `pyspi` framework (Cliff et al., 2023) to the problem of graph construction for graph neural networks on multivariate time series, which require a relational graph that is rarely given. Graph construction is the critical upstream design variable in these pipelines, yet existing methods either commit to a single pairwise statistic or learn edges from latent embeddings with no statistical semantics. We parameterise construction over a vocabulary of $K{\approx}125$ named pairwise statistics — causal, spectral, linear, information-theoretic — and jointly learn which are task-relevant, yielding an interpretable statistical signature that latent models structurally forego in favour of fleixbility. On synthetic topology classification, vocabulary-based models achieve 99.9% macro-F1 where symmetric baselines plateau at 59% and latent baselines remain at chance. The learned weight vector $\mathbf{w}$ recovers spectral Granger causality as the dominant coupling mode, aligned with theoretical construction.

1. Introduction

Message-passing neural networks (MPNNs) propagate information along relational structure (Gilmer et al., 2017; Battaglia et al., 2018). For multivariate time series (MTS), this structure is unobserved and must be constructed from data. The construction choice is the critical design variable: Han et al. (2026) showed that message-passing *consistently degrades* when graph construction is naive, and Liu et al. (2025) found substantial variation across 239 pairwise statistics benchmarked on the same neuroimaging data. Yet in every existing GNN pipeline, this choice is made once by the practitioner, before learning begins.

Current practice follows two regimes, which are limited in scope. The first fixes a graph from a single statistic and commits to one notion of dependence, i.e., Pearson correlation for functional connectivity (Bullmore & Sporns, 2009), transfer entropy for causal graphs (Duan et al., 2023), spatial distance for traffic (Li et al., 2018). So, if the task-relevant coupling is directed, nonlinear, or frequency-specific, the operator becomes the bottleneck. The second learns the graph from latent embeddings (Kipf et al., 2018; Wu et al., 2019; 2020), which recovers greater flexibility but sacrifices semantics. Even the most semantically structured latent approach (NRI's discrete edge types from (Kipf et al., 2018)) are learned categories with no pre-specified statistical meaning, and attention-based methods (Sriramulu et al., 2023) that initialise from statistics overwrite them during training. In scientific domains where the graph is the object of study — neuroscience, climate, genomics — this absence of statistical semantics is a limitation.

The intuition behind why this choice matter can be developed by considering three VAR(1) motifs: chain ($A{\to}B{\to}C$); fork ($A{\leftarrow}B{\to}C$); and collider ($A{\to}B{\leftarrow}C$). Chain and fork are Markov-equivalent (Verma & Pearl, 1990), that is, they share the same skeleton and notable V-structures, so any symmetric statistic assigns them identical edge weights regardless of sample size. No MPNN on such a graph can exceed $2/3$ accuracy on the three-class problem. Directed statistics — where $f(X_i, X_j) \neq f(X_j, X_i)$ — break this equivalence.

We propose a third regime which interpolates on the fixed-statistic—latent-embedding spectrum: parameterise graph construction over a *vocabulary* of K named pairwise statistics, computed via `pyspi` (Cliff et al., 2023), spanning six complementary families: causal (e.g., transfer entropy, Granger causality, spectral GC — *asymmetric*), spectral (coherence), linear (covariance), information-theoretic (MI), distance, and rank-based metrics. A learned weight vector $\mathbf{w} \in \mathbb{R}^K$, jointly optimised with the MPNN, identifies which statistics are task-relevant. This follows Veličković et al. (2022), i.e., rather than re-learning what "connectivity" means from raw data, we supply structured prior knowledge about pairwise dependence and let end-to-end learning select from it. The learned $\mathbf{w}$ is itself a scientific output, or a named, testable hypothesis about which form of coupling drives the task.

2. Method

Given $X \in \mathbb{R}^{M\times T}$, we compute K statistics per ordered channel pair via `pyspi` (Cliff et al., 2023), yielding a descriptor tensor $\mathbf{E} \in \mathbb{R}^{M\times M\times K}$. Graph construction is a linear probe of this descriptor space:

$$A_{ij} = \mathrm{softplus}\big(b + \mathbf{w}^\top \mathbf{E}_{ij}\big)\,, \quad \text{top-}d \text{ sparsification per node} \tag{1}$$

with $K{+}1$ learnable parameters. The linearity is deliberate (cf. Alain & Bengio, 2017), i.e., if the vocabulary is well-structured, a linear function suffices and $\mathbf{w}$ remains directly

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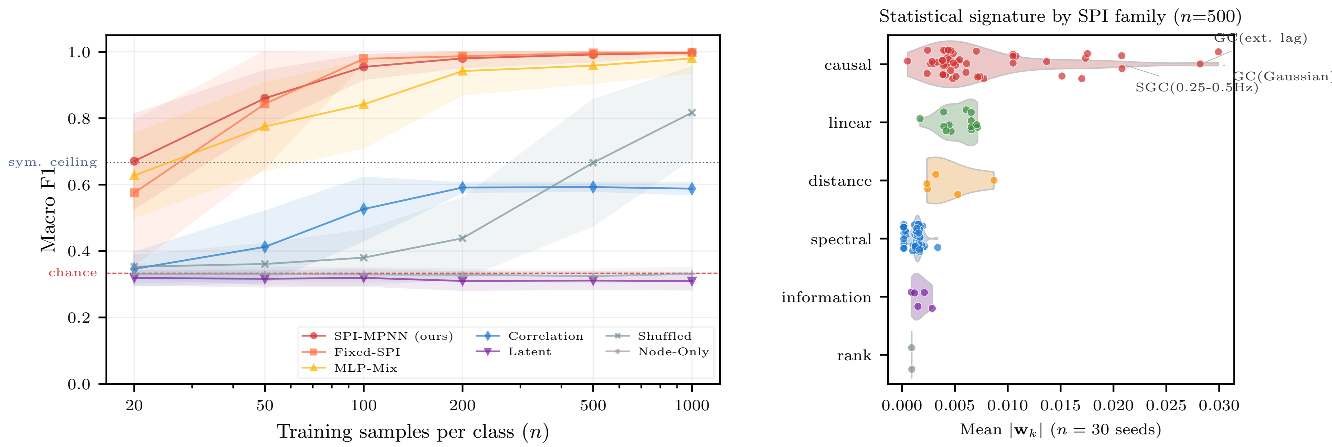

Figure 1. (a) Vocabulary-based models reach near-perfect F1 from n=100 while Correlation plateaus below the $2/3$ ceiling and Latent remains at chance. Red dashed: $1/3$ chance; grey dotted: $2/3$ symmetric ceiling. (b) Family L_2 norms of learned $\mathbf{w}$ at n=500: the causal family dominates, with individual dot weights revealing spectral Granger causality as the top-weighted statistic.

interpretable. Group lasso regularisation (Yuan & Lin, 2006) over the six SPI families encourages family-level selection in the loss-optimisation landscape:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_1 \|\mathbf{w}\|_1 + \lambda_g \sum_{g \in \mathcal{G}} \|\mathbf{w}_g\|_2, \tag{2}$$

where retained edges carry the full descriptor $\mathbf{E}_{ij}$ as attributes. An edge network $\phi(\mathbf{E}_{ij})$=MLP$(\mathbf{E}_{ij})$ conditions messages on the statistical content of each edge:

$$\mathbf{m}_{ij}=\text{MLP}(\mathbf{h}_j \| \mathbf{h}_i \| \phi(\mathbf{E}_{ij})), \quad \mathbf{h}_i'=\mathbf{h}_i+\text{LN}\Big(\textstyle\sum_j A_{ij}\, \mathbf{m}_{ij}\Big) \tag{3}$$

The SPI vocabulary therefore plays a dual role in determining *which edges exist* (via $\mathbf{w}$) and *what information flows along them* (via ϕ). Global pooling produces graph-level predictions. Training: Adam, LR warmup (60 epochs), cosine decay, restart selection (best of 2 initialisations per seed).

3. Experiments

Setup. M=10 node VAR(1) processes (T=500) with one of three directed motifs (chain, fork, collider) embedded among 7 nuisance AR(1) channels. Coupling $\alpha \sim U(0.3, 0.7)$; 1500 instances/class; K=125 after filtering; 30 random seeds. Note that chain and fork motifs produce *identical* symmetric pairwise statistics by design with chance accuracy equal to 33%.

Table 1. Macro F1 (%, 30 seeds, ±1 s.d.. Full: Appendix A).

n/cls	SPI	Fixed	E-Abl.	Corr.	Latent	Node
20	**67**±14	58±22	40±15	35±5	32±2	33±2
50	**86**±8	84±19	42±16	41±11	32±2	33±2
100	**98**±4	98±1	58±26	53±9	32±2	33±2
500	99±2	**99**±.3	93±11	59±1	31±2	32±2
1000	**100**±.3	**100**±.3	96±7	59±2	31±2	33±2

The vocabulary is the contribution, not the architecture. All SPI-consuming models outperform non-SPI baselines. Fixed-SPI, which feeds the vocabulary as dense edge features on a fully connected graph with no learned topology, matches SPI-MPNN, which confirms that the representation of pairwise dependence matters more than the mechanism used to construct edges from it. Notably, the oracle-best single statistic (spectral GC) achieves 87% at n=100 but with seed instability (std 22%, vs SPI-MPNN's 4%); at n=20 it falls to 36%, below Correlation. The full vocabulary reduces the impact of these failures (std $\leq$2% at $n\geq$50) through the diversity of 125 complementary estimators — i.e., many noisy views of pairwise dependence, when combined, provide a stability edge through a higher-dimensional descriptor vocabulary which can capture richer structures within the data.

Symmetric baselines confirm the theory. Correlation (59%) never exceeds the $2/3$ Markov-equivalence ceiling. The latent model (cf. Wu et al., 2019) remains at chance (31–32%) across all sample sizes. That is, node-level embeddings without statistical priors cannot access the chain–fork distinction. These are expressiveness limitations of symmetric graph construction.

An interpretable statistical signature. Under group lasso, the causal family (48 SPIs) carries $\sim$3.5× the L_2 norm of the next family (Figure 1b). The five highest-weighted individual SPIs are all spectral and time-domain Granger causality variants in the 0.25–0.5 Hz band (Appendix B), matching the generating process's causal structure (Geweke, 1982). An edge-ablation study (same topology, zeroed edge features) drops F1 to 93±11%, supporting the MPNN's use of the statistical content of edges in addition to the topology (see Appendix A).

Though shown on a synthetic task, this is the form of output that flexible latent graph learners forego and fixed-statistic methods may miss due to coarse-graining: a falsifiable claim which has a semantically meaningful, statistical interpretation. Pending access approval to the TUH-EEG Seizure Corpus, we look to apply this method to clinical EEG, where the question shifts from “can we recover known causal structure?” to “what form of inter-channel dependence predicts seizure onset?”.

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A. Full Ablation Results

Table 2 provides macro F1 for all model variants. Each isolates a specific component of the method.

Table 2. Full results (macro F1 %, 30 seeds). SPI = SPI-MPNN, Fixed = Fixed-SPI, Mix = MLP-Mix, SGC = SGC-Only (oracle-best single SPI), E-Abl = Edge-Ablation, Corr = Correlation, Lat = Latent, Shuf = Shuffled, Node = Node-Only.

n/cls	SPI	Fixed	Mix	SGC	E-Abl.	Corr.	Lat.	Shuf.	Node
20	67±14	58±22	63±10	36±9	40±15	35±5	32±2	35±3	33±2
50	86±8	84±19	78±7	80±18	42±16	41±11	32±2	36±6	33±2
100	95±4	98±1	84±7	87±22	58±26	53±9	32±2	38±8	33±2
200	98±3	99±1	94±5	98±7	75±24	59±1	31±3	44±12	33±2
500	99±2	100±.3	96±5	98±5	93±11	59±1	31±2	67±19	32±2
1000	100±.4	100±.3	98±1	98±5	96±7	59±2	31±2	82±14	33±2

The progression from single statistic to full vocabulary isolates the source of the method's advantage.

Node-Only (33%, chance). No pairwise features — per-channel summary statistics carry no motif information. Pairwise structure is necessary.

Latent (31–32%, chance). Per-sample adjacency from learned node embeddings, symmetric by construction. Cannot access the chain–fork distinction even with unlimited data. This is an expressiveness failure, not a training failure.

Correlation (35–59%). Fixed graph from Pearson $|r_{ij}|$. Symmetric, so provably limited to $\leq 2/3$. The gap between ceiling and observed 59% arises because top-d sparsification selects high-correlation nuisance pairs over lower-correlation motif edges.

Shuffled (35–82%). SPI values randomly permuted across pairs. Near chance at small n; rises to 82% at n=1000 via distributional fingerprinting. Confirms that pair-correspondence — not merely the marginal distribution of SPI values — is necessary for sample-efficient learning.

Edge-Ablation (40–96%). Same learned topology, zeroed edge features. The 10-point gap and 3× higher variance at n=500 confirm that edge content is load-bearing.

SGC-Only (36–98%). The oracle-best single SPI reaches 98±7% at n=200 but with catastrophic seed failures at intermediate sizes (std 22% at n=100, vs SPI-MPNN's 4%). At n=20 it falls to 36%, below Correlation. A single-statistic adjacency provides insufficient gradient signal for reliable convergence; the full vocabulary eliminates these failures through estimator diversity.

MLP-Mix (63–98%). Nonlinear adjacency offers no advantage over the linear $\mathbf{w}$, validating the linear probe design.

Fixed-SPI (58–100%). Fully connected graph with $\mathbf{E}_{ij}$ as dense edge features. Matches SPI-MPNN at $n \geq 100$. The vocabulary provides sufficient information for relational reasoning even without learned topology; the advantage of SPI-MPNN is interpretability via $\mathbf{w}$.

B. Top Learned Weights

Under group lasso (λ_g=0.02, λ_1=0.001), the five highest mean $|w_k|$ across 30 seeds at n=500:

SPI	Family	$\lvert\bar{w}_k\rvert$
SGC parametric mean (0.25–0.5 Hz, order 1)	Causal	0.0226
GC Gaussian (k=1, l=1)	Causal	0.0225
GC Gaussian (extended lag)	Causal	0.0206
SGC parametric max (0.25–0.5 Hz, order 1)	Causal	0.0202
SGC parametric mean (0.25–0.5 Hz, free order)	Causal	0.0141

All five are directed temporal measures. For a VAR(1) generating process, causal structure manifests as asymmetric spectral transfer functions in the 0.25–0.5 Hz band — the model recovers this from data. The linear and distance families retain moderate weight (Figure 1b), consistent with their role as general coupling detectors, while information and rank families are driven near zero by the group lasso.